5.13.3 Constructing a parallelogram when its two diagonals and one included angle are given

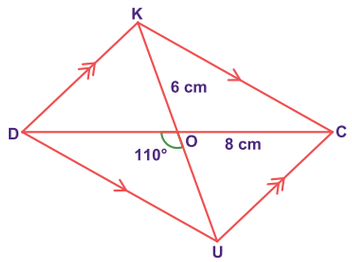
Example 5.32

Construct a parallelogram DUCK with DC=8*cm*, UK=6*cm* and $\angle DOU = 110°$. Also find its area.

Solution:

Given:

DC=8*cm*, UK=6*cm* and $\angle DOU = 110°$



Rough diagram

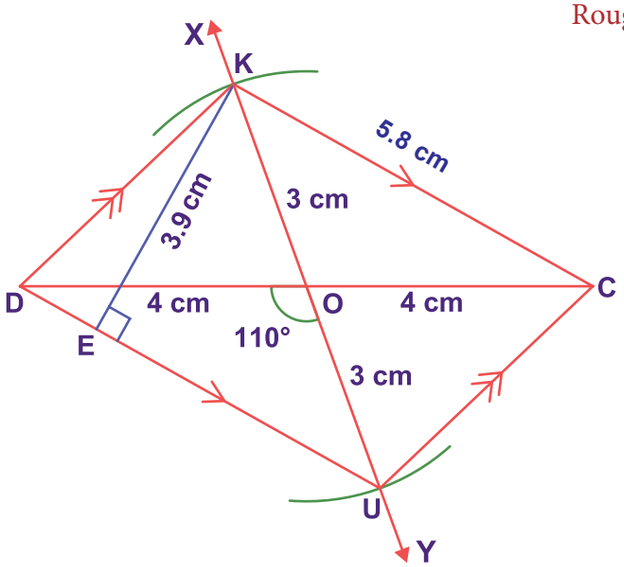


Fig. 5.51

Steps:

1. Draw a line segment DC=8*cm*.
2. Mark O the midpoint of $\overline{DC}$.
3. Draw a line $\overleftrightarrow{XY}$ through O which makes $\angle DOY = 110°$.
4. With O as centre and 3*cm* as radius draw two arcs on $\overleftrightarrow{XY}$ on either sides of $\overline{DC}$. Let the arcs cut $\overrightarrow{OX}$ at K and $\overrightarrow{OY}$ at U
5. Join $\overline{DU}, \overline{UC}, \overline{CK}$ and $\overline{KD}$.
6. DUCK is the required parallelogram.

Calculation of Area:

Area of the parallelogram DUCK = bh sq.units

$= 5.8 \times 3.9 = 22.62\, sq.cm$

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